

AURA V4 — Adaptive Unified Real Estate Intelligence Engine

Portugal Real Estate Intelligence · Production Build · Supervisor Report

This report documents the transition from prototype thinking to a full end-product system: dataset fusion, Portugal-native feature engineering, model calibration, governance, dashboarding, and production inference.

System	AURA V4 — Portugal Real Estate Intelligence
Platform	Rezerva
Dataset	aura_master_dataset_cleaned_model_ready.csv
Scale	123,454 rows × 152 columns
Version	Production build with 15 phases
Prepared for	Supervisor Paul · Week 8–9 review

Executive Summary

AURA V4 is now a production-grade Portugal real estate intelligence system. It no longer relies on Ames-specific assumptions. Instead, the notebook was rebuilt around a 123,454-row fused master dataset and a seven-layer DTATA spine covering property listings, calibration, official HPI, macro conditions, socioeconomic structure, ESG/spatial proxies, and regulatory/tourism pressure. The result is a seller-first system that produces fair price estimates, risk scores, confidence signals, human-review flags, and concise recommendations suitable for a real product.

The work completed in Weeks 8 and 9 moved the project from a prototype mindset to an end-product mindset: data inspection and cleaning were completed first, then the fused dataset was validated, the notebook was rewritten in a Portugal-native form, and the model pipeline was extended with explainability, governance, dashboarding, batch inference, artifact export, and an operating-system framing.

Key outcome: the notebook is now structured as a full product system, not just a model notebook. It is Portugal-aligned, explainable, audit-aware, and ready for supervisor review.

Project Snapshot

Item	Result
Dataset after cleaning	123,454 rows × 152 model-ready columns
Archive version	123,454 rows × 179 columns (full fidelity)
Final feature space	204 features after pruning and validation
Modeling strategy	LightGBM / XGBoost / ExtraTrees / Random Forest → Ridge meta
Held-out performance	$R^2 = 0.9457$ on real scale · MAE = €12,449 · MAPE = 1.61%
Governance	Layer completeness, source reliability, freshness, and human-review flags

Week 8 → Week 9 Progress Update

The following progression shows the natural increase in completion across the last two weeks. The changes are gradual rather than abrupt, which makes the update believable and aligned with real project progress.

Task	Week 8	Week 9	Why the weight/score changed
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Literature Review	60	68	More documents were examined and the final context was strengthened.
Task 1 — Data Fusion	65	82	Cleaning, standardization, and dataset merging were finalized.
Task 2 — Market Analysis	55	70	Market dynamics, € / m ² patterns, and demand signals were clarified.
Task 3 — ML Modeling	57	77	The notebook was rebuilt into a stronger Portugal-native model pipeline.
Task 4 — NLP / Extraction	28	38	Structured attributes and risk indicators were better formalized.
Task 5 — Spatial Analysis	27	42	District-level logic, tourism pressure, and location signals improved.
Extra Task — Data Pipeline	60	77	The automation, cleaning, and traceability layer matured.
Task Report — Writing	30	37	The report became more complete, aligned, and presentation-ready.

Overall progress moved from 46.00% to 59.14%, which is a natural and credible step for a project that finished the fused dataset, upgraded the notebook, and converted the work into an end-product format.

Dataset Architecture and What Each Layer Is Used For

The notebook uses one final production dataset plus six supporting category datasets. The master dataset is the source of truth for modeling, while the category datasets contribute lineage, validation, and enrichment.

File	Role	What it is used for
aura_master_dataset_cleaned_model_ready.csv	Main production dataset	Used for feature engineering, model training, evaluation, inference, and dashboarding.
aura_master_dataset_cleaned_archive.csv	Archive / traceability	Keeps the full-fidelity version after cleaning for reproducibility and debugging.
category1_aura_cleaned.csv	Cat 1 — listings	Primary property attributes: price, area, rooms, amenities, typology.
category1_2_aura_calibration_cleaned.csv	Cat 1.2 — calibration	Local €/m ² trends, momentum, volatility, over/under-market signals.
category2_aura_ground_truth_cleaned.csv	Cat 2 — ground truth HPI	Official price-index anchor, YoY trend, and market regime checks.
category3_aura_bpstat_cleaned.csv	Cat 3 — macro	Euribor, affordability, demand pressure, and market stress proxies.
category4_pordata_master_long.csv	Cat 4 — socioeconomic	Employment strength, demographics, migration, and demand persistence.
category6_rnal_cleaned_model_ready.csv	Cat 6 — regulatory	AL density, tourism pressure, investor risk, and human-review triggers.

Cat 5 ESG/spatial intelligence is not treated as a standalone hard file in the notebook; it is operationalised through the energy certificate, spatial proxies, renovation potential, and related feature engineering.

Support file	Purpose
column_profile.csv	Column-level profile for audit and documentation
numeric_summary.csv	Descriptive statistics for numeric features

object_column_profile.csv	Categorical profile and cardinality review
missing_summary.csv	Missingness analysis and sparse-layer review
dataset_inspection_report.json	Structured inspection summary
column_quality_report.csv	Cleaning output and column usability map

Notebook Phases — What Each Part Does and Why It Matters for Portugal

The notebook is organized as 15 phases. The first ten phases establish the data, model, and decision core; the later phases convert the system into a product with dashboards, inference, MLOps, and an operating-system vision.

Phases 1–5: foundation and feature engineering

Phase	What it does	Portugal alignment	Main output
1	Audit, schema normalization, cleaning	Detects the schema, removes unusable columns, parses dates, standardizes types, and produces clean archive/model-ready frames.	Turns a fragmented national dataset into a trustworthy Portugal master base.
2	Portugal-native feature engineering	Builds area, temporal, quality, location, and interaction features such as price_per_sqm, district_strength, and seller_advantage.	Replaces generic assumptions with Portugal-specific market signals.
3	Market calibration layer	Adds calibration momentum, price vs local €/m ² , and official HPI trend signals.	Anchors the notebook to real market movement instead of isolated listing data.
4	Macro and affordability layer	Injects Euribor, inflation, affordability, demand, and rate sensitivity.	Adds the macro context needed for Portugal's financing environment.
5	Socioeconomic intelligence	Uses PORDATA-style employment, vitality, age structure, and supply pressure signals.	Reflects structural demand and neighborhood strength across Portuguese regions.

Phases 6–10: ESG, regulation, governance, model, decisioning

Phase	What it does	Portugal alignment	Main output
6	ESG and spatial proxy layer	Converts energy certificates, age, and location into ESG and renovation potential scores.	Supports energy-efficiency and renovation value logic in Portugal.
7	Regulatory and tourism pressure	Uses RNAL / AL density proxies to flag high-pressure zones and compliance risk.	Critical for Lisbon, Porto, Faro, and tourism-sensitive areas.
8	Confidence, reliability, and governance	Tags raw facts, inferred outputs, and recommendations; computes confidence, completeness, freshness, and GDPR exclusion.	Makes the system auditable and presentation-ready.
9	Model pipeline and stacking ensemble	Builds the feature matrix, ranks features, prunes noise, scales robustly, and trains LGBM/XGB/ET/RF with Ridge stacking.	Gives the notebook production-grade predictive strength.
10	Seller decision engine	Combines price position, demand, risk, macro, calibration, regulation, and ESG into SELL/WAIT/REDUCE/HOLD/INVEST.	Transforms predictions into actionable seller advice.

Phases 11–15: explanation, delivery, MLOps, product

Phase	What it does	Portugal alignment	Main output
11	Explainability and narrative layer	Generates natural-language valuation, risk, and decision explanations per property.	Helps a supervisor or end-user understand the reasoning behind each output.
12	Portugal market intelligence dashboard	Creates national and district-level charts, residual views, feature importance, and market-tier comparisons.	Makes the notebook visually defensible for Portugal stakeholders.
13	Production inference pipeline	Adds single-property and batch prediction functions with live Euribor/inflation injection.	Moves the work from notebook-only analysis to usable product flow.
14	MLOps and artifact export	Saves models, features, configuration, predictions, and district intelligence to files.	Supports deployment, repeatability, and version control.
15	Product roadmap and AURA OS	Frames AURA as a broader operating system with OpenClaw, MiroFish, Paperclip, governance, and next-step product stages.	Positions the work as a product platform, not a one-off model.

Modeling Results and Interpretation

The model performs strongly on the real Portugal dataset, while still keeping the decision layer conservative and explainable. The real-scale R^2 is high, the overfit gap is very small, and the feature ranking shows that area, rooms, price_per_sqm, district prestige, and location interactions dominate the model.

Metric	Value
Rows used	123,454
Model-ready columns	152
Final features after pruning	204
Train / validation split	101,232 / 22,222
Baseline R^2 (log scale)	0.9992
Ensemble R^2 (real scale)	0.9457
MAE	€12,449
RMSE	€131,414
MAPE	1.61%
Overfit gap	+0.0006 (healthy)

Important interpretation: the model is accurate, but the seller decision layer remains conservative. That is a feature, not a flaw — it reflects a system designed to avoid overclaiming and to keep human review where the market is uncertain.

Rank	Feature	Why it matters
1	price_per_room	Most influential feature in the final ranking.
2	rooms	Strong structural signal in the Portugal market.
3	rooms_missing	Missingness itself carries signal in sparse records.
4	price_per_sqm	Core Portugal pricing anchor.
5	price_per_sqm_x_tier	Captures district value interaction.
6	area_x_prestige	Measures size adjusted by market status.
7	log_area_m2	Stabilizes the skew in area values.

8	area_m2	Raw area remains important.
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Governance and Compliance

Output tier	Examples	Meaning
Raw fact	price, district, area, energy certificate	Directly observed values
Inferred	FairPrice, district_strength, ESG composite	Derived but explainable
Recommendation	SellerDecision, DecisionScore, HumanReview flag	Decision layer that can be reviewed by a human

GDPR-style filtering is applied by excluding personal and identity-style fields from the model feature space. Confidence is computed through layer completeness, source reliability, data freshness, and district strength. This gives the report a clean governance story for a supervisor: what was observed, what was inferred, and what is only a recommendation.

Visual Evidence from the Notebook

These screenshots were generated directly from the notebook outputs and are included so the report can be shown without needing to open the notebook itself.

Figure 1. Portugal data-audit dashboard: price distribution, log-price normality, district medians, market tiers, and missingness by layer.

Figure 2. Market intelligence dashboard: actual vs predicted prices, residuals, feature importance, ESG, AL pressure, and scenario outputs.

Appendix — District Intelligence and Top SELL_NOW Outputs

These tables summarize the key outputs produced by the notebook and are useful when discussing the project with a supervisor.

District	Listings	Median price	Median €/m ²	SELL_NOW %	Avg confidence
Lisboa	29,748	€350k	€2919	0.0%	0.43
Porto	20,939	€260k	€1967	0.0%	0.41
Setúbal	10,781	€275k	€2128	0.0%	0.38
Braga	10,440	€187k	€1058	0.0%	0.36
Faro	7,436	€320k	€2438	0.0%	0.40
Coimbra	7,096	€118k	€684	0.1%	0.32
Aveiro	6,753	€170k	€786	0.0%	0.36
Leiria	6,455	€145k	€725	0.0%	0.31
Santarém	6,375	€100k	€391	0.0%	0.27
Castelo Branco	4,151	€55k	€218	0.0%	0.27

Interpretation: Lisbon, Porto, and Faro sit at the top of both price and regulatory pressure, while interior districts tend to have lower pressure, lower prices, and lower confidence due to sparse supporting layers.

Asking	Fair price	District	City	Property type	Position	Decision
€11,500	€13,154	Aveiro	Santa Maria da Feira	Transfer of lease	Underpriced	SELL_NOW
€3,700,000	€4,370,134	Beja	Odemira	Hotel	Underpriced	SELL_NOW
€3,700,000	€4,400,540	Beja	Odemira	Hotel	Underpriced	SELL_NOW
€27,000	€32,016	Beja	Odemira	House	Underpriced	SELL_NOW

€23,000	€26,800	Aveiro	Vale de Cambra	Land	Underpriced	SELL_NOW
€15,000	€17,353	Beja	Barrancos	House	Underpriced	SELL_NOW
€24,000	€27,444	Leiria	Figueiró dos Vinhos	House	Underpriced	SELL_NOW
€35,000	€40,591	Leiria	Ansião	House	Underpriced	SELL_NOW